

CULVERTCRAWL · REQUIREMENTS

CulvertCrawl requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/culvertcrawl		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Laptop software outputs specified in the decided order (R5 deflection report first, R13 observation log second, FHWA ratings later; CVC-DEC-001)
0.7	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.6	2026-09-30	Amish Chadha	Status from CVC-CAL-001 v0.4 after the design for construction (CVC-DDR-003); R12 met to not met
0.5	2026-09-26	Amish Chadha	Budget approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3. Record Amish's 2026-09-25 decisions (CVC-DDR-001); R12 target \$900; status from CVC-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with first-order status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

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These requirements were checked by calculation at TRL 3 in CVC-CAL-001 v0.5 (docs/04-calcs/01-sizing.md). Version 0.6 follows the constructable design of CVC-DDR-003 (2026-09-30, open for Amish's review): no target changed, but the parts added for construction put the priced BOM at \$1,034, so R12 is over the value-engineering target by \$124 (a hypothetical control target of \$910, not a limit). Earlier status follows. After the design changes Amish accepted on 2026-09-25 (CVC-DDR-002: 0.46 kg more ballast and the laser ring plane moved to 300 mm ahead), R2 and R5 are met on paper with thin margins, and R12 was within the \$910 value-engineering target, which Amish set on 2026-09-26 to match the \$906 priced BOM (CVC-DDR-002). No requirement is not met. Targets are still to be validated with users (see CVC-PRB-001). The pipe range in R1 and R5 was decided by Amish on 2026-09-25 (CVC-DDR-001); the R12 target was first set on 2026-09-25 (CVC-DDR-001) and moved to \$910 on 2026-09-26 (CVC-DDR-002). The status column is a calculated estimate, not a measurement.

The **design case** used throughout is a straight 600 mm (24 in) corrugated steel culvert, 50 m long, on a 5 % slope, with a wet silt invert and up to 150 mm of water flowing at 0.5 m/s. Unless stated, the crawler enters at the outlet and drives uphill against the flow, the worst direction. In 150 mm of water the 106 mm crawler is fully submerged. As operating guidance (CVC-DDR-002), crews should enter flooded pipes from the upstream end where access allows; the design case keeps the worst direction.

■ Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CVC-CAL-001)
R1	Fit the target pipes	Drive in round pipes of 300 to 900 mm (12 to 36 in) inside diameter; crawler no wider than 180 mm and no taller than 140 mm	Model check in 300 mm and 900 mm pipe sections	Met (170 x 106 mm, 611 mm long; 50 mm or more clearance)
R2	Reach	Drive 50 m into the design case pipe and back, pulling the tether, with 20 % traction reserve	Traction and tether drag calculation; later pull test	Met on paper (75 m wet and uphill; 64 m dry uphill; more than 100 m downhill); about 23 m if submerged track friction is 0.45
R3	Recover without entry	Crawler can be pulled out by the tether if disabled; tether break strength 1 kN or more, at least 5 times the worst stuck-pull estimate	Tether datasheet; pull calculation	Met (about 59 N locked-track pull, factor 17; strength member on an eye bolt in the ballast plate); jamming by debris not calculable
R4	Video	1080p video at 25 frames per second or more, live at the operator and recorded, with distance and time overlay; lighting to show the far wall of a 900 mm pipe	Camera and link budget; later image review	Met on paper (25 fps from a 50 fps stream, 53 % link use, about 370 lx); Pi 4 throughput not verifiable at TRL 3
R5	Cross-section profile	Measure the wall profile in the ring plane and report mean diameter and ovality (deflection) to within 1 % of nominal diameter, every 0.1 m of travel or less; the laptop software's first output is this deflection report (decided 2026-10-02)	Error budget; later calibration pipes	Met on paper: 0.25 % at 300 mm, 0.57 % at 600 mm, 0.97 % at 900 mm (thin margin), with the boom fin's shadow; spacing 6 mm
R6	Locate observations	Distance from the pipe mouth within 1 % or 0.2 m, whichever is larger; pitch and roll within 1 degree	Encoder and IMU datasheets	Not verifiable at TRL 3 (tether slack and wheel slip)
R7	Drive in water and silt	Operate in 150 mm of water flowing at 0.5 m/s over a silt invert; survive full submersion to 1 m for 30 min (IP68 target)	Drag and buoyancy calculation; later tank test	At risk: holds in the flow on paper; IP68 not verifiable at TRL 3
R8	Climb	Hold position on a 5 % slope with power off; cross a 40 mm step or joint offset	Worm gear self-locking; track geometry	At risk: hold met; step limit between 35 and 53 mm
R9	Endurance	4 h or more of mixed driving and profiling per charge	Energy budget	Met (6.4 h nominal, 5.1 h derated)
R10	Portable and quick	Whole kit 25 kg or less, no single case over 10 kg, carried by two people in one trip; set up from vehicle to driving in 10 min or less	Mass estimate; setup sequence	Met on mass (22.2 kg; heaviest item 7.8 kg, the reel with tether); setup time not verifiable at TRL 3
R11	Electrical and laser safety	Tether and crawler 48 V DC nominal (below the 60 V DC extra-low-voltage limit), fused, with overcurrent cutoff and an emergency stop at the surface; laser Class 2 or lower under IEC 60825-1	Design review	Met by design (10 A and 2 A fuses, 1.5 A cutoff)

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CVC-CAL-001)
R12	Affordable	Full kit (crawler, 60 m tether, reel, surface box) \$910 or less in parts, operator laptop excluded (value-engineering target set by Amish, 2026-09-26; was \$900)	Priced BOM (bom/ bom.csv)	Over the value-engineering target by \$124 (\$1,034 against \$910, after the parts added for construction in CVC-DDR-003). Prices are indicative
R13	Open and repairable	Common parts; every tether and module joint pluggable; open file formats (MP4 video, CSV profiles, JSON observation log with PACP-style codes); the laptop software's second output is the observation log in NASSCO PACP vocabulary, and FHWA condition ratings come later (decided 2026-10-02)	Design review	Met by design

■ Requirements not met or at risk

- **R12 is over the value-engineering target by \$124** (estimated \$1,034 against a \$910 target). The parts that make the design buildable (CVC-DDR-003) added \$128 to the \$906 that the target was set to match on 2026-09-26. Cost drivers and savings worth trying are in the Value engineering section of the design decisions register.
- **R2 (reach) is met on paper.** The constructable crawler is 6.63 kg (3.77 kg net when submerged) and reaches 75 m driving uphill against 0.5 m/s flow (52 m in v0.5, 39 m before CVC-DDR-002). The result is still very sensitive to track friction under water: at 0.45 it falls to about 23 m. Measuring that friction is TRL 4 work and is on hold. Entering flooded pipes from the upstream end is now operating guidance.
- **R5 at 900 mm is met on paper with a thin margin.** With the ring plane 300 mm ahead (was 200 mm), the top of a 900 mm pipe is 70.1 degrees off the camera axis (was 76.5 degrees) and the vertical diameter error is 0.97 % (was 1.59 %).
- **R7 and R8 are at risk:** seals (IP68) and the 40 mm step cannot be settled on paper.
- **R6 cannot be verified at TRL 3.**

■ Assumptions

- Track-to-silt friction coefficient about 0.6 and track motion resistance about 0.15 of weight; tether-to-pipe friction coefficient about 0.5.
- Crawler mass about 6.6 kg with ballast (3.77 kg net when submerged); tether about 55 g/m; both are estimates until parts are selected.
- 50 m reach covers most road crossings; 60 m of tether allows 10 m from the reel to the mouth.
- A 1 % profile accuracy resolves the roughly 5 % deflection limit used for plastic pipe acceptance with a 5 to 1 margin.
- The operator's own laptop runs the viewing and profiling software. Its outputs come in this order: the deflection report (R5), then the observation log in NASSCO PACP vocabulary (R13), with FHWA condition ratings later.